The Quantized Space-Ledger — Long Technical Explainer: Narration Script

Source: The Quantized Space-Ledger, by Edmond Ying — current revised 73-page version (file updated Oct 29, 2026). Voice: en-US-AndrewNeural. A speculative proposal; standard physics and proposal are labeled throughout.

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[0:00] Introduction

0:00 The Quantized Space-Ledger

This is a technical walkthrough of a white paper by Edmond Ying, titled The Quantized Space-Ledger: Locating Quantum Non-Locality in a Discrete Spacetime Substrate. This video follows the current, revised seventy-three page version of the paper. The paper is explicit that it is speculative. It offers a potential candidate hypothesis, not an established result, and this video keeps that framing throughout.

0:25 How to watch this

Because the audience here may be skeptical, we will separate two things carefully. Standard physics, meaning results that are measured or derived in mainstream theory, will be labeled as such. The ledger proposal, meaning the paper's own reinterpretation or hypothesis, will be labeled as a proposal. Where the paper says something is an open problem, we will say so too.

0:49 Roadmap [roadmap]

We will cover twelve chapters: the client and server picture, now with a finiteness principle; the photoelectric effect; mass as frequency, with a candidate hardware picture; measurement as queries, with numerical tests of alternative read-out rules; entanglement as a shared record, with a toy Bell test; the information cap per node; the Higgs field versus the Higgs boson; a toy lattice study of the critical point; steps toward gravity, including light bending and Mercury; black holes without singularities; testable predictions; and the open problems.

[1:25] Chapter 1 · The Client–Server Picture

1:25 Chapter 1

Chapter one. The Client–Server Picture.

1:29 The core hypothesis

The paper's single organizing hypothesis is this: spacetime itself is a discrete, entangled network of quantized nodes, called the Space-Ledger, and that network carries the quantum state. In other words, the wavefunction is assigned to spacetime rather than to the particle. What we call a particle is the localized excitation that is rendered on the client side when the ledger is queried by a measurement-like interaction.

1:55 Two layers

The framework uses two layers. The server side is spacetime, the ledger. It holds the global quantum state, the phases, the entanglement, and the Bell correlations, and it operates at a scale approaching zero length, the Planck regime. The client side is the observable universe: matter, signals, detectors, records, and observers. It is local. Nothing on the client side propagates faster than light, massive excitations move slower than light, and massless ones move at light speed. The paper stresses that this analogy is a guide to intuition, not a derivation.

2:32 Why discreteness [standard physics]

Why assume discrete nodes at all? The paper cites three lines of standard physics that hint in that direction. First, black-hole entropy: the Bekenstein-Hawking result, S equals k A over four l P squared, read as a bound, caps the information in a region by its boundary area. Second, ultraviolet divergences in quantum field theory, which a physical minimum length would cut off naturally. Third, an operational limit: probing below roughly the Planck length would take so much energy in so small a region that it would form a black hole.

2:58 No infinities [proposal: finiteness principle (hypothesis)]

The revised paper adds a working principle, labeled a hypothesis: the finiteness principle. No physical quantity in the model becomes infinite. First, nodes have a fixed size of about one Planck length, so no physical distance is shorter. Second, each node carries a finite workload, capped, in the recommended form, by a frame-independent limiting curvature. So curvature, density, and collision energy per node all have maximum values. Candidate consequences: a finite core in place of the black-hole singularity, and a natural cutoff for ultraviolet divergences. The cap value is a placeholder, not derived, and a cutoff alone does not preserve Lorentz invariance.

3:43 An organized grid [proposal: toy simulation (paper Fig. 6)]

Here the revised paper changes its emphasis. A regular grid picks out preferred directions and a rest frame. The paper now keeps an organized, body-centred-cubic grid, and argues that sums over link-only hop paths leave a direction dependence of the speed of light only at order E over the Planck energy, squared. On this cubic test grid, the hop-path front becomes round: a diagonal to axis radius ratio of 0.997 at 120 ticks. A random, causal-set-like network is kept as a fallback if boost invariance requires it.

4:18 The speed limit [proposal (built to match standard c)]

On the network, the ledger updates in discrete ticks of roughly the Planck time, on nodes spaced roughly the Planck length apart. Because the update couples only neighboring nodes per tick, nothing advances more than one node per tick. That gives a client-side speed limit of c equals a over tau, which matches the Planck length over the Planck time by construction.

4:40 Locating non-locality

Bell tests, which agree with quantum mechanics, show that any framework reproducing quantum predictions must be non-local in Bell's sense somewhere. The ledger does not eliminate that non-locality. It locates it: an entangled pair is one server-side state. Each client-side read looks locally random, so the no-signaling condition holds, and the correlation becomes visible only when records are compared over an ordinary channel limited by c. The paper notes a remaining tension: if the read-out is a physical update, it needs an ordering of spacelike-separated events that clients cannot observe. That is open. Chapter five returns to this with a toy test.

[5:21] Chapter 2 · Light in Quantized Packets

5:21 Chapter 2

Chapter two. Light in Quantized Packets.

5:24 The photoelectric effect

This chapter is standard physics background. When light strikes a metal, electrons can be ejected. The classic observation is that whether electrons come out depends on the light's frequency, not on its brightness. Below a threshold frequency, no electrons are emitted no matter how intense the light. Above it, electrons appear immediately, even at low intensity, and brighter light produces more electrons, not faster ones.

5:52 Ledger reading

How does this connect to the ledger? In the paper, a particle is not an autonomous object sliding across a continuous plane; it is a localized excitation of the network. On that reading, a photon arriving at the metal is an excitation of the ledger that is registered, or rendered, as a single localized quantum when it interacts with an electron. To be clear, this is an interpretation of the same standard physics. It changes no prediction about the photoelectric effect.

[6:21] Chapter 3 · Mass as Frequency

6:21 Chapter 3

Chapter three. Mass as Frequency.

6:24 Compton frequency [standard physics → ledger reading]

Combine two standard relations. Rest energy is E equals m c squared, and quantum energy is E equals h f. Setting them equal gives each massive particle a characteristic frequency, the Compton frequency, f C equals m c squared over h. This is standard physics. The paper's step is to read rest mass, in ledger terms, as a characteristic frequency of an excitation pattern on the network.

6:53 Mass as per-hop processing

The paper also gives mass a role in propagation. In client-server language, mass is extra per-hop processing. At every tick, a massive excitation has some amplitude to flip its direction of travel, its chirality, instead of simply moving on. This follows Feynman's checkerboard model and the Dirac quantum walks and quantum cellular automata in the literature. A massless excitation hops one node per tick and travels at exactly c. A massive one zigzags, so each hop is still at c, but the average drift is slower.

7:27 The update rule [known quantum-walk results, used by the proposal]

In one spatial dimension, the excitation is a two-component amplitude, right-moving and left-moving. One tick mixes them with an angle theta, where sine theta equals m c squared tau over h-bar. The exact lattice dispersion relation is cosine omega tau equals cosine theta times cosine k a. In the continuum limit this becomes the Dirac equation, with E squared equals p squared c squared plus m squared c to the fourth, and the group velocity is c times the square root of one minus m c squared over E, squared.

8:04 A candidate hardware picture [proposal: toy simulation (paper Fig. 7)]

The revised paper also offers one candidate hardware picture, explicitly not a claim about how the server is built. Each node is an identical core, fixed on a body-centred-cubic lattice with eight neighbours. At every tick, cores act at once on non-overlapping pairs in a checkerboard, or Margolus, pattern, so an amplitude moves at most one core per tick. In a one-dimensional run of

300 ticks, probability outside the light cone was exactly zero, packet speeds matched the dispersion formula to within 0.013, and probability was conserved to about ten to the minus thirteen. Naive in-place updates, by contrast, wipe data, leak probability, or break the light cone.

8:49 Speed depends on mass relative to energy

So speed depends on mass relative to energy, not on mass alone. The more of an excitation's energy is rest energy, the slower it runs. As energy greatly exceeds m c squared, speed approaches c, and when E equals m c squared, the excitation is at rest. Special relativity then emerges at low energies, but is not built in exactly; deviations would appear at order E over the Planck energy or higher, depending on the graph.

8:54 Three generations [data: standard · harmonics: hypothesis]

The paper then raises a speculative hypothesis. Matter fermions come in three generations that share charges and interactions and differ only in mass. The paper proposes that the generations may be three resonance modes, or harmonics, of one excitation pattern. For the charged leptons, the muon-to-electron mass ratio is about 206.77, and the tau-to-electron ratio about 3477.2, so their Compton frequencies stand in the same ratios. The empirical Koide relation is an example of a pattern such a model would need to reproduce. Neither the Standard Model nor the paper explains it.

9:35 Neutrino oscillation

Neutrinos show frequency structure directly. Each flavor state is a superposition of three mass states, mixed by the PMNS matrix. The phases of the mass components advance at rates that differ in proportion to the mass-squared differences, so the flavor composition oscillates with distance L and energy E, with phase proportional to delta m squared L over four E. This was established by Super-Kamiokande and SNO, and recognized by the 2015 Nobel Prize. The curve on screen is an illustrative two-flavor sketch, not fitted data.

10:11 Caveat [caveat]

Everything about neutrino oscillation just described is standard physics. The harmonic picture reinterprets it and does not derive it. The model does not currently predict the mass-squared splittings, the mixing angles, or the absolute neutrino mass scale, and it would gain real support only if those quantities followed from the network's mode structure.

[10:32] Chapter 4 · Measurement as Queries

10:32 Chapter 4

Chapter four. Measurement as Queries.

10:35 Which-way anchor [standard physics]

Start with the double slit. A which-way detector at one slit couples detector states to the paths. Because the detector states are orthogonal, the cross term at the screen vanishes, and the fringes disappear. This is standard quantum mechanics written in node language.

10:52 Step 1: decoherence

The paper splits collapse into two steps, and is explicit about which is explained and which is postulated. Step one is decoherence. The detector quickly entangles with many environment nodes, and the reduced state becomes effectively diagonal. In ledger language, the phase correlation is not destroyed. It is spread into a huge number of environment nodes, where no local client can practically gather it back. This is standard decoherence theory, restated.

11:21 Step 2: read-out

Step two is the read-out, and it is a postulate. Decoherence explains why alternatives stop interfering, but not why one definite outcome occurs. The read-out postulate states: when a client-side system extracts information from the ledger, creating an irreversible, decohered record, exactly one outcome k is sampled, locked, and returned to the client side, with probability equal to the squared magnitude of c k. That is, the Born rule is taken as the ledger's read-out rule.

11:53 Main open problem [open problem]

The paper stresses that this is a postulate, not a derivation, and calls deriving it the main open problem. The revised version narrows that problem. Gleason's theorem shows that, given a Hilbert-space structure and additive outcomes, the square is the only consistent read-out. The remaining question is why server-side amplitudes behave like rotatable arrows, that is, why the ledger's state space carries a complex Hilbert-space structure with unitary evolution. It is also open whether the read-out is a physical server-side update or only a client-relative fact.

12:27 Testing alternative read-outs [proposal: numerical check (paper Fig. 1, Sec. 3.3)]

The paper also tests alternatives numerically, with illustrative parameters: Sorkin's three-slit parameter kappa against measured bounds, merging detectors, and signalling through an entangled pair. Power rules give kappa of about five times ten to the minus three to four times ten to the minus two, exceeding both bounds. Softmax and exponential rules fail all three tests. Only the squared amplitude passes all three. This assumes additivity of the read-out, which a concrete ledger model would still have to justify.

13:01 No conscious observer needed

In this model, measurement does not require a conscious observer. Any physical interaction whose cascade of effects records information about a state in its environment acts as a query to the server, and yields a definite client-side outcome. Stated more carefully, client-observable events correspond to measurement outcomes of server-side quantum evolution. This is the ledger reading of decoherence and of Zurek's quantum Darwinism.

13:27 Einstein's Moon

The paper uses an example from Abraham Pais. Einstein asked whether the Moon exists only when someone looks at it. The Moon is continuously scattered by sunlight, by cosmic microwave background photons, and by other particles. So in ledger terms it is queried continuously, and remains definite whether or not anyone observes it.

13:49 Caveat and Zeno [caveat]

The caveat matters. Decoherence accounts for the suppression of observable superpositions, but it does not by itself select a single outcome, and the paper remains neutral among interpretations, such as Copenhagen and many-worlds, on how that selection is understood. A related example is the quantum Zeno effect, observed by Itano and colleagues in 1990: frequent queries keep re-committing the same state and suppress evolution. The paper calls this a reinterpretation that leaves the standard calculation unchanged.

14:21 Query-rate limits [standard physics + one proposal (last item)]

The revision adds query-rate limits. Signals and records travel no faster than light. The Margolus–Levitin bound allows at most two E over pi h-bar orthogonal states per second. And, as a model assumption with no direct experimental support, each node performs at most one update per Planck time, about ten to the forty-three per second.

[14:45] Chapter 5 · Entanglement as a Shared Record

14:45 Chapter 5

Chapter five. Entanglement as a Shared Record.

14:49 The pointer interface [proposal: shared record (hypothesis)]

Section 5.5 writes the server-side picture as a minimal interface. A call, entangle, opens one shared record; each daughter packet carries only a pointer to it, plus its conserved share of energy and momentum. Relays forward the packets as ordinary link phase. A call, read, runs on the server: it computes the odds, locks the attribute for the whole set, and returns plus or minus one. No outcome is stored in advance; a pre-stored value would be a local hidden variable, capping S at two.

15:23 Toy Bell test [proposal: toy simulation (paper Fig. 4)]

With standard CHSH angles, the shared record gives S equals 2.822 plus or minus 0.007, against the quantum maximum of two root two, about 2.828. With eighty percent detector efficiency and one percent dark counts, it gives 2.800 plus or minus 0.009. A control that gives each daughter its own copied record gives 0.01 plus or minus 0.01. Each side alone is fifty-fifty, and the near side's statistics do not shift with the far side's setting. The record is non-local, but no message travels faster than c.

16:03 Caveat [caveat]

The caveat is central. The read inserts the cosine-squared odds by hand, so this toy checks that the interface is consistent; it does not derive the Born rule. Deriving those odds from the node rules is listed as an open problem, as is how a grid of local cores carries out the joint read without client-detectable signalling.

[16:23] Chapter 6 · Entangled Space & the Information Cap

16:23 Chapter 6

Chapter six. Entangled Space and the Information Cap.

16:28 The network as a tensor [standard formalism, ledger interpretation]

Formally, the paper models spacetime as a graph whose vertices are nodes and whose edges are entanglement bonds. Each node carries a finite-dimensional Hilbert space, and the state of a region is a network tensor, in the spirit of matrix product states or MERA. The mathematics is standard quantum mechanics in node language; the novel content is the interpretation and the discreteness hypothesis.

16:53 The bound [standard physics]

Now the information cap. The Bekenstein-Hawking entropy of a black hole is A over four l P squared, in natural units. Converted to bits, that is about one bit per four ln two, roughly 2.77 Planck areas, or about 0.36 bits per Planck area. Under the holographic principle, no region can hold more information than a black hole of the same boundary area.

17:19 Applied to cubic nodes

The paper applies this to cubic Planck nodes carrying one qubit each. A single cube has six faces of one Planck area, so its cap is about 2.2 bits, and one qubit fits. But for a block N nodes on a side, independent qubits would give N cubed bits, while the surface cap is only about 2.2 N squared. The volume count exceeds the cap once N is larger than about 2.2.

17:45 Bits per node [proposal (applied bound)]

So inside a block N nodes wide, each node can carry at most about 2.2 over N independent bits: about 0.2 for N equals ten, and about ten to the minus twenty for a proton-sized region, where N is around ten to the twenty.

18:02 Interpretation

The interpretation is that each node can still hold a qubit, but node states within a region cannot all be independent. Nearly all must be entangled with one another or determined by the boundary, so free degrees of freedom scale with surface area rather than volume. In ledger terms, the server stores a region largely through its boundary, with a tightly correlated interior. This is consistent with holography and with proposals, by Van Raamsdonk and by Maldacena and Susskind, that spacetime connectivity arises from entanglement.

18:35 Caveats [caveat]

The caveats: the argument assumes the holographic bound applies to the lattice, which has not been proven for a discrete network. A simple cubic lattice is anisotropic, so the factor 2.2 depends on node shape, although the N squared versus N cubed scaling does not. And the model does not yet supply an update rule that produces area-law entanglement.

[18:58] Chapter 7 · Higgs Field vs Higgs Boson

18:58 Chapter 7

Chapter seven. Higgs Field versus Higgs Boson.

19:02 Field and boson

In ledger language, the Higgs vacuum value, v, about 246 GeV, is a property of the network's ground state, held server-side at every node. The Higgs boson, observed at the LHC in 2012 and now measured at about 125.2 GeV, is a client-side excitation of that state: a ripple on the ground state. The measured values are standard physics; the server and client assignment is the proposal.

19:32 Stiffness of the ground state [standard physics]

At tree level, the Higgs mass equals the square root of two lambda, times v. That gives lambda of about 0.13, and a ratio m H over v of about 0.51. The paper reads this ratio as a measure of how stiff the ground state is against small displacements.

19:51 The hierarchy problem [open problem]

The hierarchy problem: radiative corrections to the Higgs mass parameter grow with the square of the cutoff, while the Higgs mass over the Planck mass is around ten to the minus seventeen. A discrete network supplies a physical cutoff, so corrections are finite. But the paper is clear that this is shared by any lattice regularization and does not remove the fine-tuning. The model offers no explanation of this hierarchy, or of the weakness of gravity, at present.

20:20 Edge of stability [published literature]

There is published precedent for near-criticality. Froggatt and Nielsen in 1996 proposed two degenerate minima of the effective potential, one near the Planck scale, and predicted a top mass of 173 plus or minus 5 GeV and a Higgs mass of 135 plus or minus 9 GeV. Shaposhnikov and Wetterich in 2010 argued that asymptotically safe gravity drives lambda toward zero near the Planck scale, giving about 126 GeV. Buttazzo and colleagues in 2013 showed the measured parameters place the vacuum close to the stability boundary.

20:59 The paper's rerun [paper's first-pass rerun]

The paper reports an independent rerun using two-loop Standard Model running with Buttazzo matching conditions. Imposing lambda at the Planck mass equals zero gives a Higgs mass of about 128.8 GeV, against the measured 125.2, agreement within about three percent. Exact agreement would need a top mass near 171 GeV versus a measured 172.6, with roughly one GeV of interpretive uncertainty in the pole-mass translation. With current inputs, lambda reaches a minimum of about minus 0.011 between ten to the seventeen and ten to the nineteen GeV: the vacuum sits near a critical point and is metastable.

21:44 What this does and does not claim [caveat]

In ledger terms, this is a candidate statement that the network's ground state sits near a critical point. It is not yet a derivation. And the paper notes that the near-zero lambda at the Planck scale is the strongest part of this discussion, but it was established in the earlier literature, not by the ledger model.

[22:02] Chapter 8 · A Toy Lattice

22:02 Chapter 8

Chapter eight. A Toy Lattice.

22:05 The toy model [toy model]

To see what near-criticality means on a discrete substrate, the paper studies a four-dimensional hypercubic lattice at Planck spacing, with eight nearest neighbours per node and one complex doublet per node. The doublet has four real components, the same content as the Higgs doublet, and matching a qubit's two complex amplitudes. The action has a hopping parameter kappa and a lattice self-coupling lambda L.

22:30 Free critical point [study figure: mean field]

In the free theory, the critical point is exactly kappa c equals one eighth, the inverse of the number of neighbours. This mean-field figure from the study shows m H over v versus kappa for several couplings, with kappa c equals one eighth marked, the value 0.49 marked, and the target ratio of 0.51.

22:51 Monte Carlo shifts [proposal: toy Monte Carlo (paper Fig. 12)]

Monte Carlo runs on lattices of size eight, twelve and sixteen locate the critical point from Binder-cumulant crossings. Self-interaction shifts it upward: kappa c of 0.1259 plus or minus 0.0005 for lambda L of 0.001; 0.1288 plus or minus 0.0009 for 0.01; and 0.1442 plus or minus 0.0014 for 0.05. In this figure from the paper, the Binder curves cross above one eighth. Mean-field theory places the critical point below one eighth for these couplings, and misses this shift.

23:33 Past the tipping point [study figure: broken phase (MC, L=16)]

Just past criticality, the measured vacuum value follows v squared proportional to kappa minus kappa c, as expected near a critical point. This figure shows the order parameter versus distance past the critical point, and the Monte Carlo estimates of m H over v.

23:50 Homing the knob [proposal: toy-tested hypothesis (paper Fig. 13)]

The revision adds a toy-tested hypothesis: self-tuning the knob by homing. A local rule can bring kappa near kappa c without hand tuning, if it has a signal that peaks there, here an anisotropy cost. Like CNC homing to a limit switch, the knob is dithered, steps at full rate toward the better side, and shrinks its step only on a confirmed sign flip. On an O four toy on an eight-link BCC grid, with kappa c about 0.1495, all twelve runs locked, from hot and cold starts, about 0.006 above kappa c.

24:27 The hierarchy, restated [toy-model result]

Here is the sobering number. To obtain v equals 246 GeV with Planck spacing requires kappa minus kappa c of order ten to the minus thirty-four. The paper is explicit that this is the hierarchy problem restated on the lattice, not a solution. In the pure Standard Model, radiative symmetry breaking does not supply the offset, because the top quark contribution dominates with the wrong sign.

24:53 Caveats [caveat]

The caveats are substantial. The lattices are small and the error bars sizable. Gauge fields and the top quark are not included. A hypercubic lattice breaks rotational and Lorentz symmetry. Mean-field and Monte Carlo disagree on the direction of the shift, and the conversion from lattice units to GeV is schematic. For the homing rule, the residual bias of about 0.006 does not shrink from lattice size eight to twelve, the rule still has hand-set constants, and getting from ten to the minus three down to ten to the minus thirty-four is not addressed.

[25:29] Chapter 9 · Steps Toward Gravity

25:29 Chapter 9

Chapter nine. Steps Toward Gravity.

25:32 Rigid nodes, varying workload [standard GR (weak field) · ledger reading is a proposal]

None of what follows is derived from a node rule yet; these are candidate paths and worked examples. The nodes are rigid. Mass-energy does not bend them; it changes the workload of nearby nodes, and so the cost of routing a signal and the rate of local clocks. In weak-field general relativity, the metric can be read as a refractive index, n of r, about one plus two G M over r c squared, an optical-metric picture going back to Gordon in 1923.

26:02 Planck-unit bookkeeping [standard identity · applied by the proposal]

The key identity: G M over r c squared equals mass in Planck masses times inverse distance in Planck lengths. The Sun holds about 3.3 times ten to the 131 Planck-scale nodes, so its mean workload is about 2.7 times ten to the minus ninety-four Planck masses per node. Each share is tiny; the effect comes only from the sum.

26:25 Solar light deflection [proposal: worked example (paper Fig. 15, Sec. 6.14)]

The Sun was coarse-grained into up to about a million cells, with masses following Eddington's standard polytrope model, and the deflection of a ray grazing the limb was summed cell by cell. It converges to four G M over R c squared, 1.751 arc seconds, with a relative difference of about ten to the minus nine. The paper is explicit that this is expected by construction. The factor of two, equal time and space contributions, is still to be derived from an explicit node rule; that is open problem twenty.

26:59 Mercury, node-summed field [caveat: PPN gamma, beta inserted (paper Fig. 17, Sec. 6.14)]

Section 6.14 applies the same node-summed solar field to Mercury's orbit. This figure shows the perihelion direction drifting over two hundred orbits, and the simulated advance against the post-Newtonian factor, two plus two gamma minus beta, over three; the dashed line is 42.98 arc seconds per century times that factor. The paper's caveat for this figure: gamma and beta were borrowed from the post-Newtonian equations of general relativity and inserted, not derived.

27:31 Clock relay [proposal: simulation (Sec. 6.18) · gamma = 1 assumed]

Section 6.18 goes further, without a figure. If a node's clock rate is relayed between neighbours, and each hop compounds the slowdown rather than adding to it, the time part of the metric becomes e to the minus two U, giving beta equal to one. In simulation, the compounding relay gave 42.91 arc seconds per century, against about 43 observed. An additive relay gave 50.1, which the paper excludes. A strictly neighbour-relayed clock produced compounding without it being imposed, with beta reaching 0.9992 on the finest grid. Gamma equals one is still assumed, not derived.

28:12 Hiding the rest frame *[proposal: toy simulation (paper Fig. 19, Sec. 6.15)]*

A rigid grid defines a rest frame. Following the Lorentz–Poincaré reading presented by Bell, a preferred server frame may exist yet stay undetectable. Two toys test this. Waves on a cubic lattice slow by a direction-dependent amount of order k a squared. And here, sine-Gordon kinks re-plotted on fixed nodes contract by exactly one over gamma, with energy gamma m within 0.04 percent up to 0.95 c, while much wider than the spacing. Near the spacing, they radiate, slow, and pin.

28:47 A new constraint *[caveat: toy result, open challenge]*

The conclusion, for these update rules: direction dependence and departures from Lorentz contraction both fall off as the pattern size over the spacing, to the minus two. That yields a new constraint: physical particles must be much larger than one node. A proton spans about five times ten to the nineteen Planck lengths, so for accessible particles the effect is below about ten to the minus thirty. The paper flags the remaining challenge: radiative corrections in virtual loops could carry grid anisotropy down to low energies.

[29:21] Chapter 10 · Black Holes Without Singularities

29:21 Chapter 10

Chapter ten. Black Holes Without Singularities.

29:25 Exterior unchanged, finite core *[proposal: finite core (hypothesis)]*

Far from the core, the network is assumed to reduce to general relativity, so horizons, Event Horizon Telescope shadows, and LIGO and Virgo ringdowns are unchanged. Inside, with a minimum node size, curvature is bounded at roughly one over the Planck length squared, and density at roughly the Planck density, about five times ten to the ninety-six kilograms per cubic metre. The singularity is a candidate to be replaced by a finite, maximally loaded core. Precedents include the big bounce of loop quantum cosmology, regular black holes of Bardeen and Hayward, Planck stars, and Markov's limiting density.

30:05 Node saturation *[proposal: placeholder cap (Sec. 6.17)]*

Section 6.17 turns this into a rough numerical estimate, using a placeholder rule that no node carries more than about one Planck mass. A mass M then cannot be packed below a minimum radius, R min. General relativity's Kretschmann curvature grows as one over R to the sixth, without limit. With the cap it levels off at about 842 over the Planck length to the fourth, the same for every mass, because the saturated density is always about the Planck density.

30:36 Curvature levels off *[proposal: toy (paper Fig. 20)]*

This figure shows the levelling-off for the Sun, one kilogram, and one Planck mass. The saturated radius is about 4.5 times ten to the minus twenty-three metres for the Sun, far inside its three-kilometre horizon. For the Sun as it is, the change from general relativity is around ten to the minus ninety-three, so the light-bending and Mercury examples are unchanged. In this toy, a horizon forms only above about 0.17 Planck masses, about four times ten to the minus nine kilograms.

31:09 Caveats [caveat]

The caveats: the cap value is a placeholder, not derived from the node rules; the minimum black-hole mass is toy-dependent; and the fate of a fully loaded core, including bounce or white-hole outcomes, is left open, as is how information escapes in Hawking radiation, following the Page curve.

[31:29] Chapter 11 · Testable Predictions

31:29 Chapter 11

Chapter eleven. Testable Predictions.

31:32 Photon dispersion [proposal: testable]

A framework that only restates quantum mechanics cannot be tested. The ledger's distinct physical claim is discreteness at roughly the Planck scale. On a regular lattice, high-energy waves feel the grid, and a generic linear modification makes photon speed depend on energy. Two photons emitted together a distance D away, with energy difference delta E, would arrive separated by delta t of about xi times delta E over the Planck energy, times D over c.

32:02 Existing constraints

Observations already constrain this. From GRB 090510, Fermi found no evidence of Lorentz-invariance violation, with a lower limit above 1.2 times the Planck energy on a linear effect. From GRB 221009A, the brightest burst ever recorded, LHAASO set limits around ten to the twenty GeV, roughly ten times the Planck energy. A naive regular grid with an order-one coefficient is therefore disfavored; the data require xi of about 0.1 or less.

32:38 Quadratic, not linear [proposal: lattice toy (Sec. 6.15)]

The revised paper's response has two parts. First, the lattice rule it tested produces no linear term; its leading effect is quadratic. At Planck spacing, the fractional speed shortfall is one eighth of E over the Planck energy, squared, times a direction factor. For a 31 GeV photon, like the highest-energy photon from GRB 090510, that is about 8.1 times ten to the minus thirty-seven along an axis. Second, a random, causal-set-like network remains a fallback, under which first-order dispersion is expected to vanish.

33:16 Gravity-mediated entanglement [proposal: testable (Sec. 7.4)]

The revision names an accessible test: gravity-mediated entanglement, as proposed by Bose and colleagues and by Marletto and Vedral in 2017. Two small masses, each in a superposition of two locations, interact only through gravity. In the ledger, each branch loads nodes differently, so the expectation is that entanglement would be observed. A null result, with a working experiment, would count against the model. The caveat: this is an expectation, not a derivation, and it matches other quantum treatments of gravity, so a positive result would not single out this model.

33:55 Small sources [proposal: estimate (Sec. 7.5)]

Relativistic effects of laboratory-scale sources have not been measured. The paper estimates fractional clock shifts: about 3.2 times ten to the minus twenty-four near a one-tonne tungsten block, and about four times ten to the minus twenty at the apex of the Great Pyramid from its own mass. Both lie below the roughly ten to the minus eighteen reached by today's best clocks. The node sum reproduces general relativity here, so such a test would check consistency, not separate the two.

34:25 Frequency ceiling and w = −1 [proposal: testable (Secs. 7.6–7.7)]

Two more commitments. One update per Planck time gives a clock rate of about 1.85 times ten to the forty-three hertz, and a ceiling near nine times ten to the forty-two hertz, energies of order the

Planck energy. No physical wave or particle should exceed it. And the flat per-node vacuum cost predicts dark energy with w equal to minus one exactly. The paper states that a confirmed time-varying dark energy, as hinted by recent DESI results, would rule it out.

[34:56] Chapter 12 · Open Problems

34:56 Chapter 12

Chapter twelve. Open Problems.

34:59 Open problems I [open problem]

The list of open problems has grown to twenty-seven. Among the most central: derive the Born read-out rule rather than postulate it, now narrowed to why the ledger carries a complex Hilbert-space structure; specify a concrete network tensor and update rule that reproduce relativistic quantum field theory; show that server-side ordering of updates stays unobservable to clients; and make the mass-loads-the-server idea quantitative, recovering Einstein's field equations and the equivalence principle.

35:29 Open problems II [open problem]

New items include: deriving the factor of two in light bending, and beta and gamma for Mercury, from one node rule; deriving the saturation cap and minimum black-hole mass; coupling chiral modes to the weak gauge field; and deriving the cosine-squared odds of the shared record. Section 8 adds framework objectives: a Born rule from local projection, Lorentz invariance under radiative corrections, and an anomaly-free chiral gauge sector.

35:58 Methodological note [method]

The paper closes with a methodological note. To stay falsifiable, server-side explanations are reserved for regimes where known physics fails or is silent: quantum correlations and measurement, Planck-scale structure, and black-hole interiors. Where established client-side physics already explains a phenomenon, the ledger reinterprets it but does not replace it. As the paper puts it, a framework that can attribute anything to the server would predict nothing.

[36:27] Outro

36:27 Summary

To summarize. The Space-Ledger is a potential candidate hypothesis: spacetime as a discrete, entangled network that holds the quantum state, with non-locality located server-side and a local, light-speed-limited client world. The revised paper adds a finiteness principle, toy checks of read-out rules and of a shared-record Bell test, worked examples toward gravity, a finite black-hole core, and new testable commitments. Much of it reorganizes standard physics; its distinct claims are discreteness, a read-out postulate, and those concrete commitments. Whether it survives depends on the open problems and on future data. The paper is by Edmond Ying. Thanks for watching.